Zev Averbach

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SUMMARY

Entrepreneurial, pragmatic software engineer who's in love with the web platform. Has founded a business and co-founded a startup, and has worked as a product engineer, taking web products from idea to launch in Python and TypeScript. Now going deep on coding agents and how their harnesses work under the hood.

EXPERIENCE

Vitol – Business Analyst / Senior Python Developer – Jan. 2024 - Present — Geneva, Switzerland

- Designed and built a trade surveillance platform from soup to nuts (React, AG Grid, FastAPI, SQLAlchemy, APScheduler, later Prefect) that lets analysts assess potentially problematic trade patterns, helping the firm stay compliant. Defined requirements with business stakeholders and worked with a development partner who implemented the detection algorithms.
- Built and maintained Prefect data pipelines for compliance and reporting requirements, largely ports of an older PL/SQL-based system.
- Migrated from self-hosted Prefect to a cloud instance with no service interruption and no duplicate flow runs. Built helpers for managing variables and blocks across Prefect environments and shared them with the team.
- Early adopter of agentic coding tools at the firm (Claude Code, later OpenCode), using them daily for development and automation.
- Built and shared a range of productivity tools with colleagues, most of them written with agents. Some were built for agents to use, and many were just as useful to human users.
- Built a tool that lets agents search pipeline logs to debug failures and tighten schedules based on typical data arrival times.
- Interviewed candidates for the middle office team (Python and React, assessed in a business context).
- Took part in a regular support rotation for the compliance-side middle office team.

Caterpillar, Inc. – Senior Data Engineer – Mar. 2020 - July 2023 — Geneva, Switzerland

- Supported existing ETL pipelines defined in Apache Airflow.
- Refactored multiple existing Python packages and Airflow DAGs serving the global aftermarket solutions division.
- Took over DBA duties for the aftermarket Snowflake data warehouse.
- Patched Airflow to make the web interface more intuitive for data engineers, as well as bolted on password strength checks and rotation.
- Built an Airflow UI plugin to facilitate documentation of our tables by those most familiar with it (engineers and analysts).
- Created proofs of concepts of web APIs to expose data augmentation logic, as well as front ends for consuming from it.
- Designed, implemented, and deployed automation to rebuild and re-grant Snowflake views when the underlying table changed or other unforeseen changes.
- Ran weekly Python trainings for my team in 2020.
- Started and ran weekly Python "lunch and learns" across the organization.

Averbach Transcription – Lead Developer – June 2015 - July 2023 — various

- Automated 90% of operational and administrative processes: Prior to taking this role, work assignment, invoicing, and confirmation of transcription projects were manual, and the business ran off of a spreadsheet.
- Created an automatic work assigner and a web API for processing transcribers' acceptance of transcription projects.
- Created automatic invoice generation. As of November 2018 99% of the invoices are also sent automatically without the need for human review. Implemented SQLAlchemy and Freshbooks API v2.
- Created a CLI for administering one-off business tasks, such as changing the due date or rate of a given project, combining invoices, adding PO numbers, etc. (Python)
- Built "slice and dice" product for qualitative research customers, which enables viewing all the answers to a given research question across many separate transcripts on a single page. (PHP)
- Pushed a handful of frequent customer requests to self-service fields on the customer interface, thus eliminating a significant portion of help desk tickets. (PHP and jQuery)

Recurse Center – Participant – Sept. 2018 - Oct. 2018

- Learned React.js and ES6 from scratch in order to build the prototype of Transcribely (see "Projects").
- Built a to-do list email bot.
- Dove into advanced Python features via pair programming with very curious, new programmers.

Tappt – Co-Founder, Full Stack Developer – May 2015 - Jan. 2021

- Formed a small team to collaboratively build the prototype of a web app that enables keyword search of YouTube videos. (Python, Flask, JS)
- Launched a commercial product based on the prototype in early 2018. Rewrote most of the app in a remote two-person team: Added user sign-up and authentication, integrated with the Stripe API for payments, and built SQLAlchemy models to replace raw SQL.
- Implemented indexing of the videos using YouTube captions, which reduced Tappt's operational costs to nearly nothing.

Averbach Transcription – Founder, President – Jan. 2008 - May 2015

- Founded Averbach Transcription to serve media and qualitative research clients.
- Known for exceptional transcript quality and rapid turnarounds.
- Recruited and managed a distributed team of remote transcription contractors, as well as a series of freelance developers and designers to first create the platform, then add features for transcription project management and assignment, client self-service, and scalability.
- Set up processes for CRM, accounts payable and receivable.
- 10% average annual growth in revenue
- 99.5%+ transcript accuracy consistently achieved
- "Slice and dice" in-browser transcript product delivered to qualitative research clients.

SKILLS

TECHNICAL COMPETENCIES: Python, JavaScript, TypeScript, FastAPI, SQLAlchemy, Prefect, Apache Airflow, Snowflake, React, AG Grid, Svelte, HTML, CSS, PostgreSQL, MySQL, Redis, Memcached, AWS, Git

SPOKEN LANGUAGES: English (native), French (B1)

PROJECTS

Kind Chess · Jan. 2023 - Current

- Wrote and deployed a complete front and back-end web application (websockets + REST) in vanilla JavaScript and Python.
- Currently refactoring to make it scalable using AWS services.

avt-fresh · Mar. 2022 - Apr. 2022

- Created an ergonomic API wrapper of Freshbooks' API v3.
- Wrote approachable documentation.

Plaid API-Integrated Web Apps · Sept. 2020 - June 2021

- Designed and built two web apps which integrated with Plaid's Assets API.
- AWS Cognito, Terraform, AWS Lambda, Render.com, Python + Flask, Svelte.js

The Syllabus · Sept. 2019 - Mar. 2020

- Architected and implemented a custom newsletter platform from soup to nuts.
- Worked around some challenging architectural choices that had already been made before I started.
- AWS SES, Python + Flask, Svelte.js.
- Launched within three weeks of starting.

Transcribe all the Things · Feb. 2019 - July 2019

- An open source CLI for submitting recordings to various speech-to-text web APIs and managing the resulting transcripts.
- Python w/ Boto3 and Google Cloud libraries

Transcript Processing · Oct. 2018 - July 2019

- An open source CLI for converting the output from various speech-to-text web APIs to a couple of standard formats.
- Python w/ Click

Viral Overlay · Feb. 2019 - Mar. 2019

- An open source CLI for overlaying styled text on a video based on the words spoken in the video.
- Python w/ Click and ffmpeg

Transcribely · Sept. 2018 - Sept. 2020

- A human-machine transcription app written in React.js + Python/Flask
- Enables a human to very efficiently edit machine-generated transcripts.
- Saves more than 50% of labor compared to a human transcribing the same recording from scratch.

EDUCATION

University of California, Berkeley · May 2003
Bachelor of Arts, History, 2003